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IDFC2026 Submission 5

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Submission 5

Title	Digital Forensics in Cybersecurity: Techniques for Effective Evidence Investigation
Paper	 (May 13, 10:22)
Author keywords	Digital Forensics Cybersecurity Network Forensics Incident Response Forensic Automation Packet Analysis Evidence Preservation Threat Detection Cybercrime Investigation Artifact Extraction
EasyChair keyphrases	digital forensics (70), computer science and engineering (60), associate professor department of computer science (51), institute of engineering and technology chitkara (51), university institute of engineering and technology (51), engineering and technology chitkara university punjab (51), computer science and engineering chitkara university (51), preservation and analysis of digital evidence (51), part of a connection initiation attempt (51), data source summary (47), degree of bachelor of engineering (46), technology chitkara university punjab india (46), chitkara university institute of engineering (46), identification collection preservation and analysis (46), digital evidence (40), forensic techniques (40), digital forensic (40), web history (40), python script (40), legal integrity and authenticity (40), port scanning attack (31), access to restricted (31), user content suspected (31), recently deleted files (31), systematic port scanning (31), future scope (30), disk analysis (30), destination ports (30), shell bags (30)
Abstract	In an era characterized by an increasing reliance on digital systems, the importance of robust cybersecurity measures has never been higher. Modern society's rapid shift toward the internet for nearly every aspect of life has been accompanied by a rise in complex cyber threats, such as data breaches, malware attacks, and unauthorized access to restricted systems. Digital forensics plays a critical role in this landscape by facilitating the identification, collection, preservation, and analysis of digital evidence required to investigate these cybercrimes. This research provides a deep dive into advanced methodologies and industry-standard tools—specifically Autopsy and Wireshark—to improve the efficiency and accuracy of digital investigations. The study incorporates a practical dimension by simulating cyberattacks, such as systematic port scanning using Nmap, within a controlled virtual environment to generate realistic forensic artifacts. A significant focus is placed on the use of automation through Python scripting (using the Scapy library) to manage the challenges posed by high data volumes and the increasing complexity of digital evidence. Traditional manual investigations are often slowed down by these factors, making timely response difficult. The findings demonstrate that integrating a structured forensic framework with automated analysis significantly enhances the



Submitted	speed and accuracy of evidence processing while ensuring the essential legal integrity and authenticity of the data.
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